

for almost 200 years and used to tap a large amount of waste heat in the industry. What works in the factory, though, is not so easy to reproduce in the human body. The small temperature differences found around our body generate tiny voltages, which are difficult to step up to useful levels, and the power output is very low.

Researchers have managed to make a thermoelectric generator that is only 1 cm² in size and can generate about 150 microwatts at a temperature difference of just 0.9°C—a difference that is easy to find around the body. The device uses thin films in nanoscales of thermoelectric semiconductor, which means that it can be made very small. These are so thin that they are barely visible.

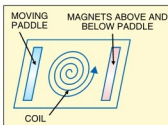


Fig. 6: An in-body microgenerator where the body movement moves a metal coil in a magnetic field, inducing an electric current

However, there is a snag that has come from the need to step up the low voltages from this material to the sort of level required for running pacemakers and other devices. Losses during conversion slash 150 microwatts scavenged to 70 microwatts

of usable power, but that is still sufficient to power a pacemaker. By connecting thousands of nano thermogenerators in series to boost the power output, it is possible to make such a small device (Fig. 5).

Another type of thermoelectric generator which has been developed for medical applications cannot be termed nanogenerator but implantable microgenerator. It provides power for an implanted electronic pacemaker that triggers the heartbeat of patients suffering from tachycardia.

A tiny radioisotope is used for heat generation, which eliminates the need for repeated operations to replace the batteries normally used. Its life is about ten years.

Microgenerator harnessing body movement

An in-body microgenerator developed last year is powered either by the heartbeat or limb movements. It works by harnessing the inertia of a moving mass either to force the charged plates of a capacitor together or to move a conducting coil through a magnetic field. The arrangement is shown in Fig. 6. The microgenerator is about 6mm across and 20mm long and can generate about 150 microwatts of output power, which can easily run a cardiac pacemaker or biosensor. The body movement moves a metal coil in a magnetic field, inducing an electric current. ●

The author is a regular contributor to EFY

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